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| wipro \| PARI | **Title** | **Surface Treatment - Plating Process Selection w.r.t. it's Properties** | Document No | PRD-GD-DES-12 |
| | | | Version No | 2 |
| | | | Revision No | 4 |
| | Classification | Reference - Guideline | Effective Date | 10-Aug-23 |

Selection of Typical Surface Treatment process-Plating on ferrous and Nonferrous material shall be specified on drawing with reference to below guideline.

Surface Treatment Type, It's properties -

| Sr. No. | Type of Surface Treatment | Properties | Input surface pre-requisites | Process Constraints | Application Notes | Application part Families | Film Thickness (in microns) | | Salt Spray in Hrs. | Hardness | Material | | | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Min. | Max. | | | Ferrous | Non Ferrous | Non Metallic | |
| 1 | Hard Chrome Plating | 1.Hard chrome plating has good wear resistance under sliding or rolling contact (when done on case hardened steel).<br>2.It has good corrosion resistance. | 1.It requires Pre and post Grinding.<br>2. All the sharp corners edges shall be blended by radius and not by chamfers.<br>3. Surface must be case hardened or base material must be through hardened ( Min 45 HRC) | 1. Plating is electro deposition the plating layer not formed on inside dimeters, corners.<br>2. Plating layer is not uniform | 1. under sliding or rolling contact (when done on case hardened steel).<br>2.Suitable for close tolerances parts, achieves harder surface.<br>3.When using for newer applications , prototype and performance test on a test coupon is recommended before mass deployment | Shafts, Rollers, Tooling components(Customer component touching parts) | 10 microns | 500 microns | 48 Hrs. Red Rust | 1100 HV | Steel Alloys | Copper, Brass Can be done but no recommended for engineering usage. Evaluate thoroughly if such plating is necessary | NA | |
| 2 | Flash Chrome Plating | 1.Flash Chrome Plating has good wear & corrosion resistance.<br>2.It has good aesthetics qualities. | 1. It requires Pre - Grinding Post grinding is not suitable.<br>2. All the sharp corners edges shall be blended by radius and not by chamfers.<br>3.If by geometry any parts not possible to grinding shall be buffed uptill machining or deep pitting marks vanishes.<br>4.Plating thickness allowances shall be considered before final sizing of the parts. | Since Plating is electro deposition the plating layer not formed on inside dimeters, corners. | 1.Flash chrome plating typically used for horizontal surfaces which are subjected to wear and tear due to repetitive placement of hand tools.<br>3.It is not suitable for rolling contact. | Shafts, Rollers, Tooling components(Customer component touching parts), tool handles, posts and pillars used in tooling, end plates and caps, table tops. | 10 microns | 20 microns | 48 Hrs. Red Rust | 510 HV | Mild Steel, Carbon Steel, Cast Iron | Copper, Brass | NA | |
| 3 | Electroless Nickel Plating (E.N.P.) | 1.ENP has good wear and corrosion resistance<br>2.It can be performed on Ferrous, Non ferrous metals as well as plastics.<br>3.It can be used in wet condition.<br>4.It has Food grade quality. | 1.It has required Pre grinding.<br>2. If by geometry grinding not possible then machining marks blended with Buffing.<br>3.Plating thickness allowances shall be considered before final sizing of the parts. | 1. For the Heat treated parts De-hydrogen embrittlement process shall be done.<br>2. For better adhesion grinding portions shall be glass bead blasting required. | Use when higher corrosion resistance required over flash chrome plating, use when chrome plating is not preferred due to statutory requirements, use when surface has intricate shapes and cavities. | Stationary parts for which no rubbing actions.(Pallet Base plates, stationary pegs etc.) | 10 microns | 15 microns | 72 Hrs. Red rust | 950 HV | Mild Steel, Carbon Steel, Cast Iron | Copper, Brass, Aluminium | Plastic (ABS) and Other | |
| 4 | Iron Zinc Black Passivation (Trivalent only) | 1. Iron Zinc Black Passivation has very high corrosion resistance. . | 1.Parts shall be machined.<br>2. If any parts processed without machining shall be buffed uptill deep pitting marks vanishes.<br>3.Plating thickness allowances shall be considered before final sizing of the parts. | 1.If any parts having small diameters, blind holes shall be cleaned with air flow and baked properly uptill no chemical residuals.<br>2.Welding shall be leak proof in the case of welded rollers or any hollow section, if any chemical pours from the weld leak inside the tube will tends to form rust. | 1. However to achieve the full benefit, the parts shall be taken care against scratches during handling and assembly<br>2.It is to be used for Ferrous machined parts, all sizes and tolerances of general purpose applications. | Stationary parts for which no rubbing actions.(Pallet Base plates, stationary pegs etc.) | 8 microns | 12 microns | 96 Hrs Red Rust. | 150 HV | Mild Steel, Carbon Steel | NA | NA | |
| 5 | Self Anodizing / Silver Anodizing | Very good for Aesthetical Quality | All surfaces shall be buffed properly | Shade at the welding and heat affected zones not matched with the plain uniform surfaces. | It is exclusively suitable for Aluminium structural Material (e.g. Extrusions, Plates etc.) | All Extrusions, Pallet base plates | 10 microns | 15 microns | NA | 100 HV | NA | Only Aluminium | NA | |
| 6 | Hard Anodizing | Hard Anodizing has good wear resistance than Silver anodising. | All surfaces shall be buffed properly | Shade at the welding and heat affected zones not matched with the plain uniform surfaces. | 1.It is to be used on Aluminium parts which are subjected to wear.<br>2. For Highest wear resistance and for wet applications ENP process is recommended instead of Hard Anodizing. | Plates family | 30 microns | 100 microns | NA | 300 HV (+) | NA | Only Aluminium | NA | |
| 7 | Electro Galvanizing | 1.It has high weathering resistance required. It has high hardness as compared to painting.<br>2. (however generally these are manufactured from GI sheets directly) or the | 1.Surfaces shall be free from heavy pitting marks, deep heavy machining marks.<br>2. Plating thickness allowances shall be considered before final sizing of the parts. | 1.If any parts having small diameters, blind holes shall be cleaned with air flow and baked properly uptill no chemical residuals.<br>2.Welding shall be leak proof in the case of welded rollers or any hollow section, if any chemical pours from the weld leak inside the tube will tends to form rust. | Electro galvanizing is suitable for structural steel parts where painting or powder coating is not possible. | Electrical cable ducts, routing parts (not structural supports) like fish plates, separators, component mounting plates, Dunnage rollers etc. | 14 microns | 25 microns | 50 Hrs. - Red Rust | 250 HV | Mild Steel, Carbon Steel, Cast Iron | NA | NA | |
| 8 | Balzer Coating | High wear resistance, low friction & high working temp. properties at surface and high toughness property at core. | 1. Residual hardness after tempering shall be more than 55 HRC.<br>2. Tools must be shot peened or sand blasted after heat treatment. | Recommended Heat treatment cycle for D2 material as per below -1. Stress Relieving-Tool should be stress relieved by gradually heating to 650 to 680 degree C. hold for 2 hrs., cool slowly to 500 degree C. and then cool to room temp.<br>2. Tools should be hardened between 1040 to 1080 degree C.<br>3. Tempering - Tools must be tempered at 500 to 540 degree C. and should be held for 2 Hrs. Temper at least twice and thrice when tolerances are to be kept close. | It is recommended for cutting tools and metal working dies intended for long service life . | Gripper locating pins, Gripper compliance pins, Cone insertion tooling for piston ring....etc. | 0.5 microns | 5 microns | -- | 1000 to 1500 HV | OHNS, D2 | NA | NA | Consult with RM before selection. Do not use En-24 material for Balzer coating surface treatment (It is Plasma assisted & high temp. coating process) |
| 9 | ZAFC (Zinc Aluminium Flake Coating) | High Corrosion resistance | Plating thickness allowances shall be considered before plating | below M5 size this plating is feasible but due to less Allen head socket size the ZAFC solution layer is more hence the Allen key slippage chances are more. | All the hardwares | Hardwares | 6 | 8 | 500 Hours | Nil | 12.9 graded hardwares | NA | NA | |
| 10 | **Quench polish quench (QPQ)** Specialized type Nitrocarburizing case hardening (Three steps are involved: nitrocarburize ("quench"), polish, and post-oxidize ("quench").) | QPQ process gives excellent resistance against wear & corrosion, Scuff, Fatigue etc on ferrous components and safeguard from the destruction of the components. | 1. It requires Pre - Grinding, Post grinding is not suitable.<br>2. All the sharp corners edges shall be blended by radius and not by chamfers.<br>3.If by geometry any parts not possible to grinding shall be buffed uptill machining or deep pitting marks vanishes.<br>4.Treatment on Fully finished components. | QPQ is not recommended for case hardened parts, Hardness of 30 to 50 HRC is preferred for through harden steels. | 1. under sliding or rolling contact parts<br>2.Suitable for close tolerances parts, achieves harder surface. | For customer part touching details, the part touching details must be hardened to 60 HRC. Use tool steel for such demanding applications Like D2, A2, O2, or other superior alloys. For being pins and shafts, case hardening is acceptable with an understanding that after QPQ the case will be around 40 HRC. | 0.2 | 0.3 | 200 Hours | as per material grade - for through harden steel 45 to 50 HRC | All through hardened steel | NA | NA | |

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**Revision History**

| DCR Ref. No. | Revision No. | Revision Date | Revision Details |
|---|---|---|---|
| Nil | 2 | 06-May-21 | Replaced PARI logo by Wipro-PARI logo, Document number has "-" seperator. |
| 19/23-24 | 3 | 14-Jul-23 | i. Guideline reviewed and updated with additional columns of Properties, Input surface pre-requisites, Process Constraints, ii. IZBP process Plating thickness changed from 14-25 microns to 8-12 microns and iii. QPQ process added. |
| 01/23-24 | 4 | 10-Aug-23 | Wipro PARI logo updated |



| QMS | Global Engineering Director | Head - QMS |
|---|---|---|
| Prepared by | Approved by | Controlled by |